

2LEAD: Lunar-LIBS Elemental Analyses and Detection Instrument

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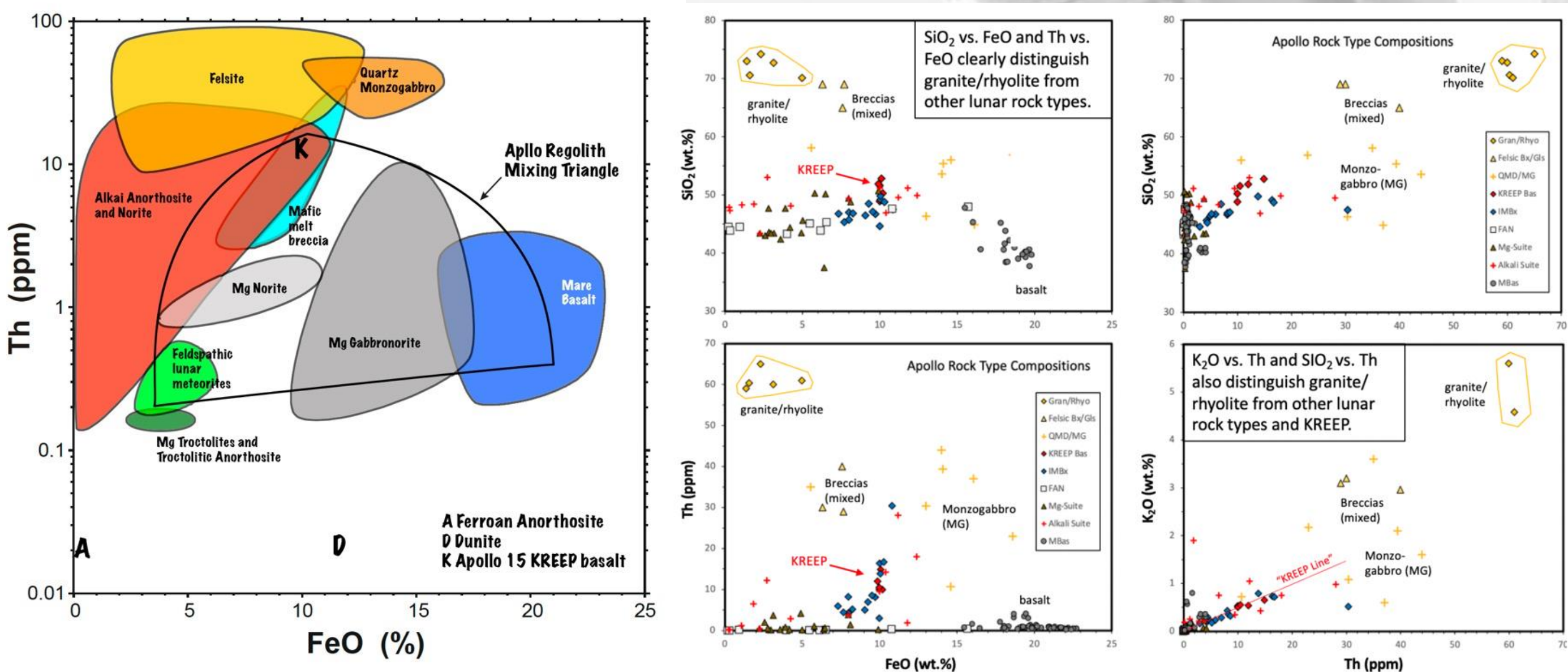
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Introduction

2LEAD is a compact Laser-Induced Breakdown Spectroscopy (LIBS) instrument for lunar surface exploration. The instrument consists of a 1064 nm laser (Lyra), focusing optics, a scannable mirror, context camera, spectrometer, and avionics computer. Named for the lyre of Orpheus, whose music was said to move stones, Lyra enables 2LEAD to probe lunar rocks and regolith from a distance (0.5–2.4 m) and reveal their elemental composition. 2LEAD supports targeted point, line-scan, and 2D analyses of features of interest, providing rapid (minute-scale), take advantage of rover or astronaut mobility. 2LEAD provides high-resolution elemental analysis of key elements (e.g., Si, Fe, Mg, Al, K, Ba) and volatiles (H as H₂O/OH⁻) in rocks, regolith, and ices. These data are critical for lunar science, resource prospecting, and habitability assessments. 2LEAD builds on the heritage of ChemCam and SuperCam but is optimized for deployment on CLPS-class mobility platforms. Its core innovation is simplicity: making LIBS practical, scalable, and deployable across cost-constrained missions, enabling routine in situ geochemistry rather than one-off flagship capability.

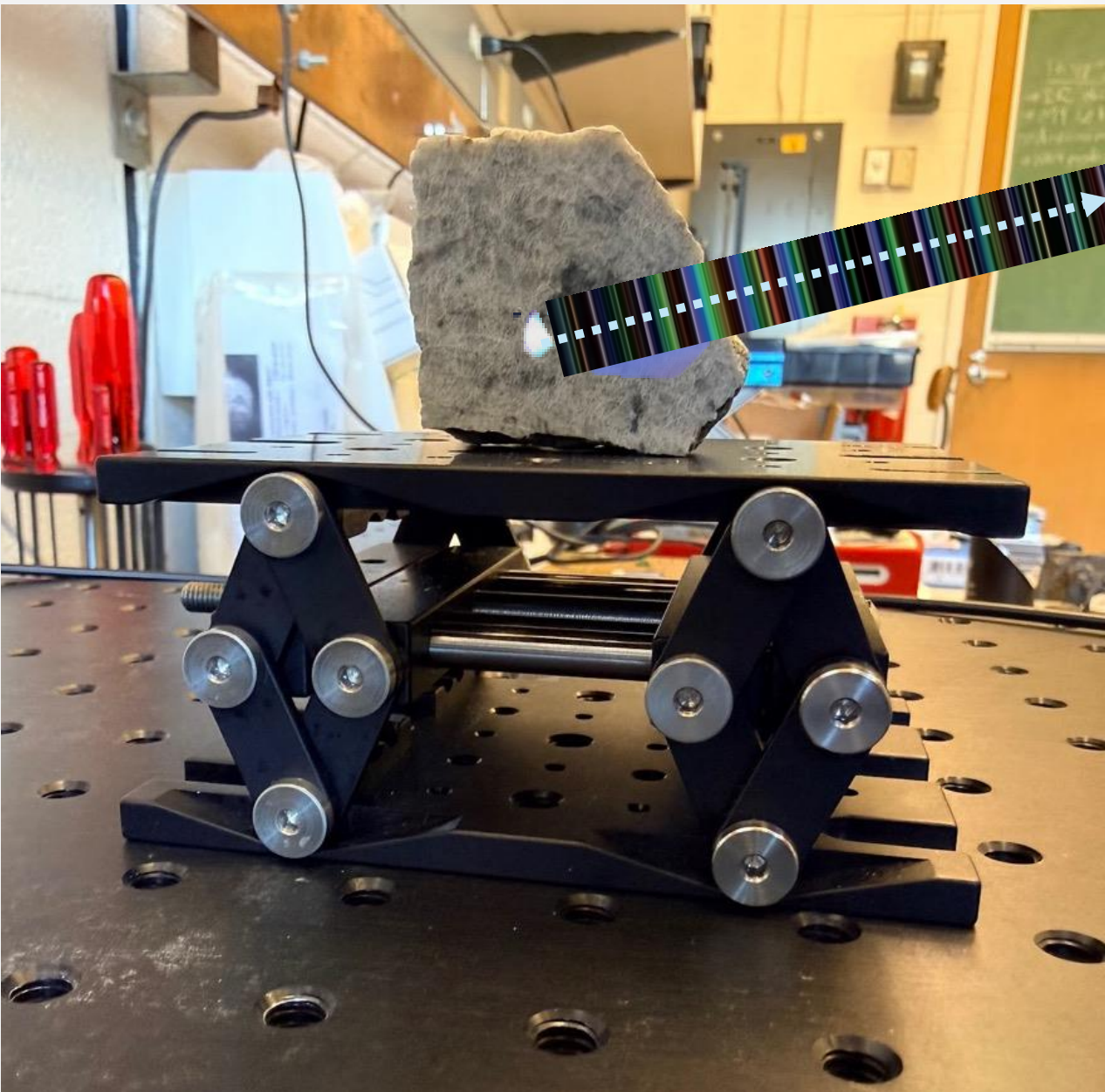
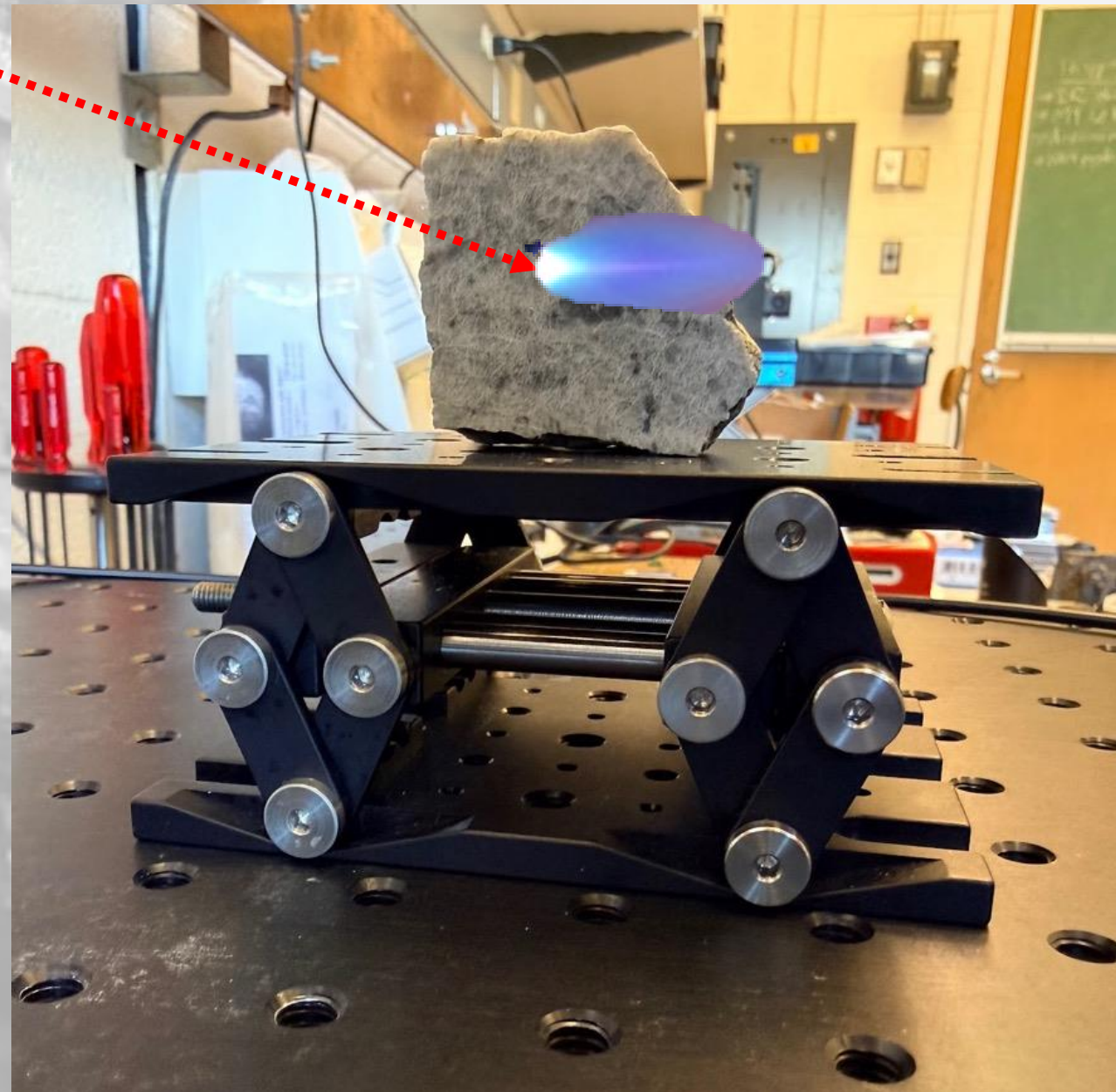
LIBS-Derived Elements Yield Lunar Rock Types



A few key elements (Fe, Ba [Th], Si, K, Ti) elements enable discrimination of various lunar rock type endmembers (e.g., ferroan anorthosite, mare basalt, felsic volcanics, KREEP basalt, etc.) and allow modelling of mixed materials.

How Does LIBS Work?

- A high-energy, short-pulse (<10 ns) laser creates a transient plasma by ablating a small volume of sample material.
 - The plasma expansion is controlled by the surrounding pressure environment.
 - LIBS has been demonstrated in vacuum and on the lunar surface (for example, Chandrayaan-3).
- Electrons in the plasma are excited to higher energy orbitals and then decay, re-emitting energy as atomic and ionic emission lines.
 - The emitted photons form an element’s signature spectrum, which is measured by a spectrometer.
 - Spectra are calibrated with standards to yield quantitative data.

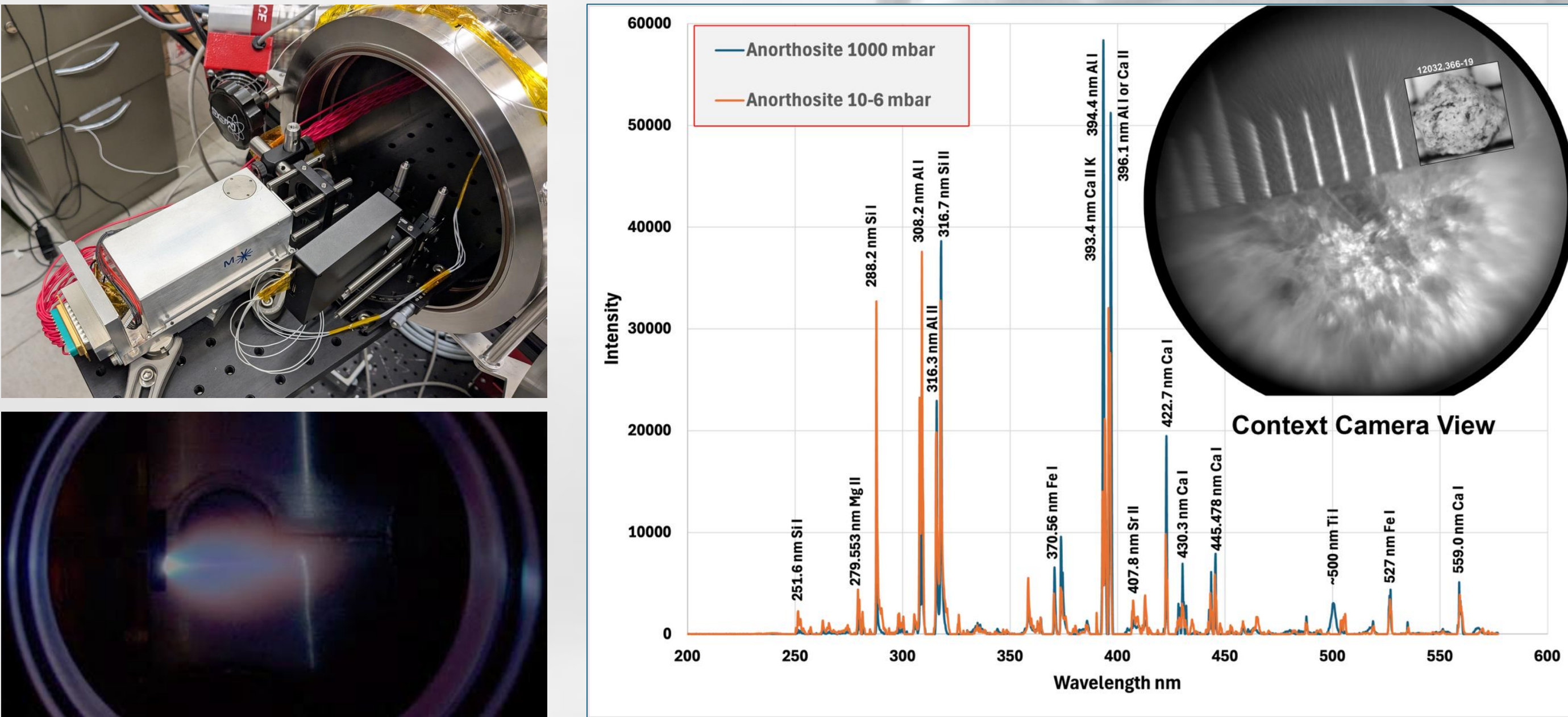


LIBS instrument performance specifications (est.)

Parameter	Value
Size	Probe: 28.6 cm × 17.2 cm × 12.7 cm (6,247 cm ³) Electronics: 12.4 cm × 15.0 cm × 7.2 cm (1,339 cm ³)
Weight (NTE)	7.1 kg
Power (Nominal Ops)	14.6 W
Power (Peak/Heating)	25.4 W
Spectral Range	180–840 nm
Spectral Resolution	0.6 nm (FWHM) or better over spectral range
Focus Range*	0.5 m to 1.5 m (extended range out to 2.4 m)
Angular Scanning*	±6° X/Y (extended range ±12°)
Imaging FOV	2.8°
Laser*	1,064 nm, <10 ns, >10 mJ, with the possibility of up to 50 mJ (extended range with active switching to control pulse energy)
Repetition Rate	Single shot, or 1-10 Hz
Operating Life	100 K samples
Time to Collect Sample	10 seconds
Operating Temp.	-20°C to 40°C
Non-operating Temperature	-40°C to 70°C
Instrument Warm-Up Time**	22.5 minutes

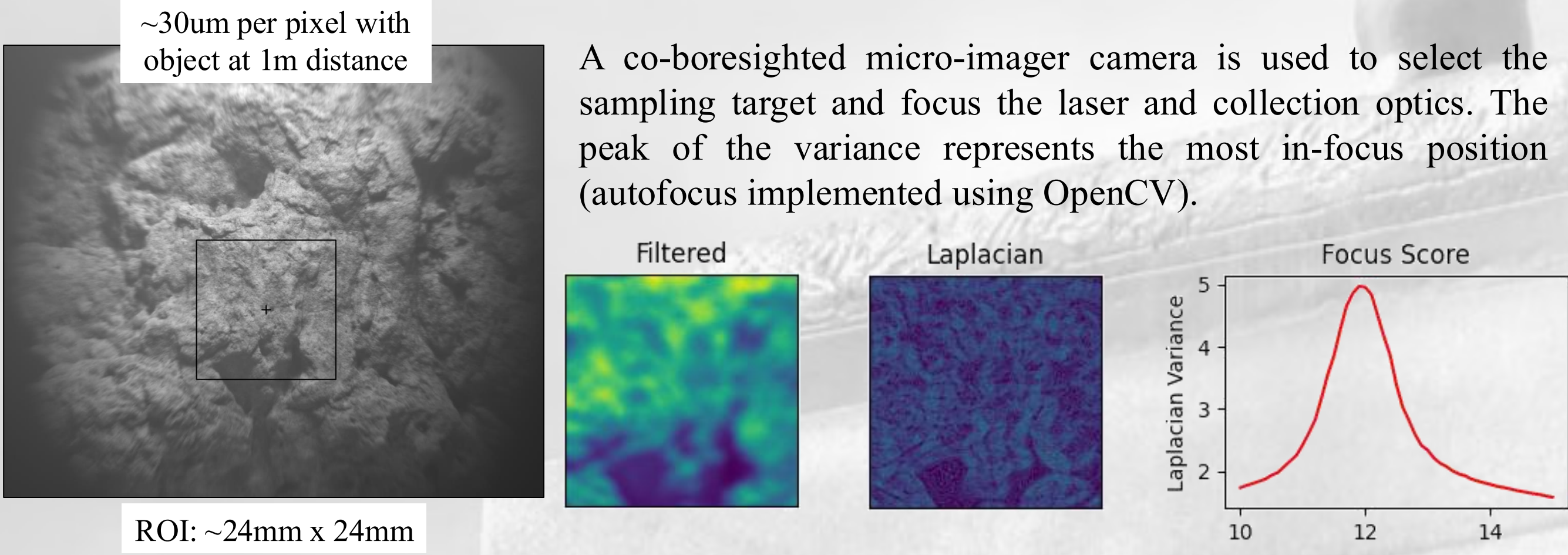
*Extended range planned to relax rover mobility requirements
**Instrument warm-up time dependent on rover boundary interface temperature, 22.5 minutes max. for interface temp of 10°C or greater

Vacuum Chamber Test Results (TRL5)



2LEAD underwent TRL-5 testing in January 2026 (top left). LIBS plasma under vacuum (lower left). LIBS spectra of an anorthosite rock show prominent emission peaks for Fe, Ca, Al, Si, and Sr at 1 atm and at 1×10⁻⁶ mbar. The inset shows a context-camera image with multiple LIBS zap pits (white marks) and a 1 mm ruler for scale. The co-boresighted camera allows robust pairing of mineral morphology and LIBS composition to assign unique chemical analyses.

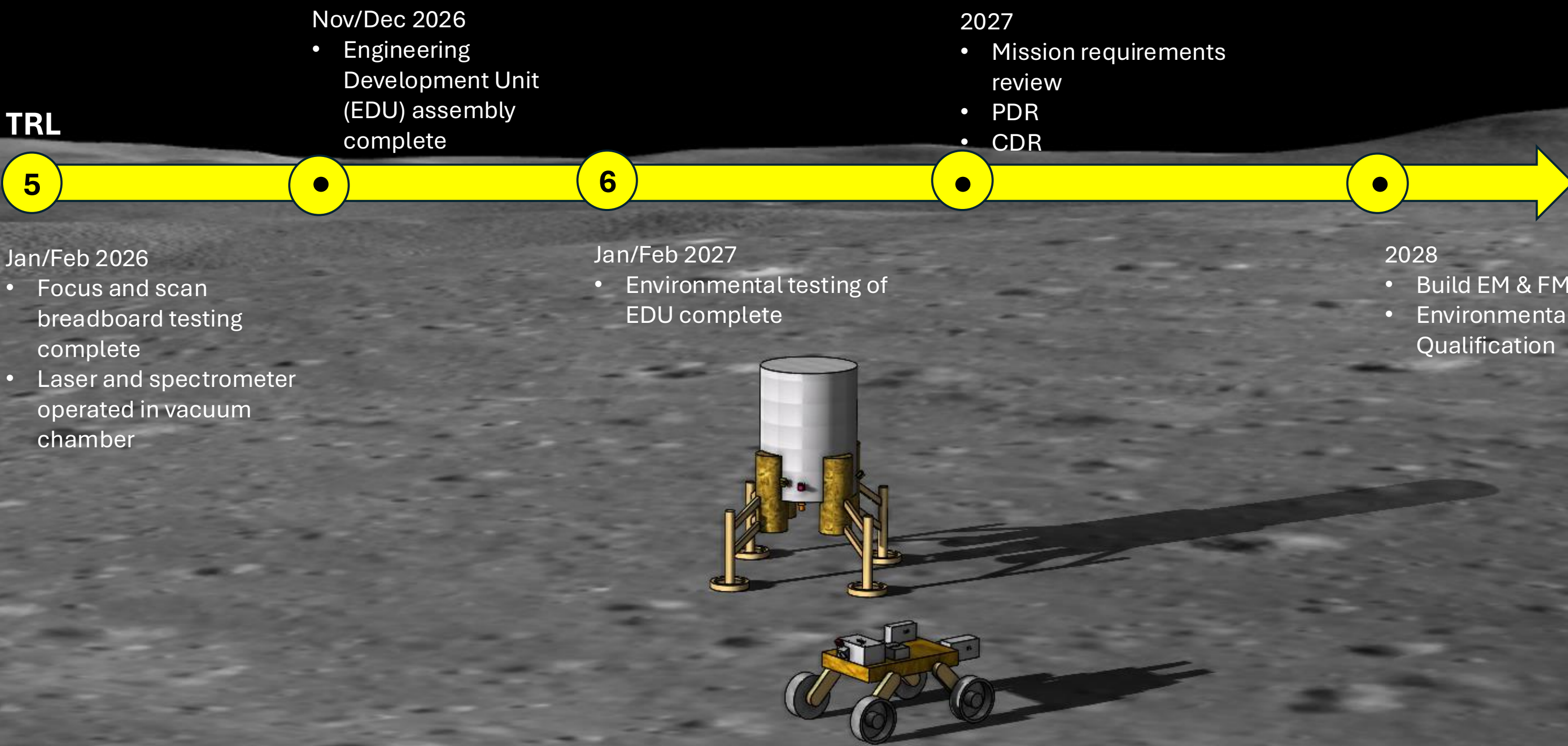
Autofocus



Mission Concept

2LEAD can be combined with complementary instruments to form a payload suite. The payload is rigidly mounted to the rover deck for thermal control, with the optical axis tilted ~30° down from the horizon to view the surface directly in front of the rover. The focus and scan mechanisms achieve a reachable workspace of 20 cm × 20 cm at a nominal working distance of 1 m. When a feature of interest has been located, an automated survey can be performed to efficiently collect a grid of LIBS samples. For example, the surface of a rock or loose regolith could be sampled at points spaced 1mm apart. Repeated shots at a single point can boost the spectrometer signal, remove dust and reveal compositional gradients. 1000s of shots can be collected over the course of a typical lunar daytime mission profile.

2LEAD Development Timeline



2LEAD prototype, Orpheus (left): Top-down view of the optical breadboard showing the 1064 nm laser (Lyra), focusing optics, X–Y scannable mirror, context camera, spectrometer, and avionics computer. The yellow line indicates the context camera optical path, the red line shows the laser beam path, and the light-green arrow traces the return path of LIBS emission to the spectrometer.

Right: front view of 2LEAD with viewport. The enclosure dimensions are 41cm L x 21 cm H x 31 cm D.

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Image generated using ACT Lunar QuickMap